Competition between radial diffusion and interfacial exchange in thermal-gradient deposition columns: conservative modeling of iodine transport

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We present a conservative finite-volume model of vapor transport and wall deposition in laminar thermal-gradient columns. Nine Pb–Bi–I gases undergo element-conserving equilibrium in a frozen helium carrier and exchange material with shared wall inventories. A Hertz–Knudsen–Schrage law acts in series with near-wall diffusion. On a Graetz–Robin reference problem, this face-flux treatment converges at second order, whereas a cell-center sink converges at first order. Fifteen baseline and 64 follow-up simulations, supplemented by production and refinement studies, distinguish numerical sensitivity from uncertain physical inputs. The maximum raw element-balance residue is 1.41×10^{-12} of supplied material. Halving the time step or adding species correctors changes iodine profiles by at most 0.323%. Upwind and bounded limited-linear advection differ by up to 20.8% at equal mesh. With limited-linear advection, the most sensitive profile changes by 1.98% between 41,472 and 93,312 cells, without a consistent observed order. Scaling estimates and upwind sensitivities suggest predominantly diffusion-controlled exchange. A $+29\,\mathrm{kJ\,mol^{-1}}$ shift in BiI gas Gibbs energy changes the BiI_3 share of deposited iodide iodine from 94.58% to 2.01%. Silica comparisons expose an unresolved temperature-coordinate inconsistency. For five published PbI_2 experiments, wall-only main-deposit onset temperatures have an RMS deviation of 18 K, compared with 35 K for two published models. This comparison is conditional on the onset region and differing carrier, vapor-pressure, and source assumptions; computed peaks remain 1–4 cm upstream. The evidence establishes conservation and verifies the wall discretization, while identifying limitations of predictive experimental agreement.

I. INTRODUCTION

Iodine released from lead–bismuth eutectic (LBE) can travel in several chemical forms before it deposits on cooler surfaces. The chemical form matters because it sets both the gas-phase inventory available for transport and the equilibrium pressure that governs wall exchange. Liu et al.[1] studied this behavior by evaporating pure iodides and iodine-containing LBE into helium, separating the products along a negative temperature gradient and characterising the deposits. Their measurements give spatially resolved evidence for PbI_2 and BiI_3 deposition, and for potassium-containing deposits in some LBE experiments. Linking such measurements to a transport calculation needs more than a single deposition temperature: the temperature-coordinate mapping, the carrier flow, the source history and the gas composition all affect the predicted distribution.

Seen as a fluid-dynamics problem, a thermal-gradient column is a slow laminar flow of an inert carrier through a long, narrow tube whose wall temperature falls by several hundred kelvin along its length (fig. 1). Trace vapors released near the hot end are advected downstream by a carrier whose density and velocity vary with the imposed temperature, and they diffuse radially towards the wall. Where the local vapor pressure exceeds the equilibrium pressure at the wall temperature, material is lost to the wall; where it falls below

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that pressure, existing deposits can evaporate again. The deposition profile is therefore set by the competition between axial advection, radial diffusion to the wall and the interfacial exchange law. This is a Graetz-type wall mass-transfer problem[2], but with a nonlinear wall condition that depends on the local wall temperature and condensed inventory. The same coupling underlies the interpretation of thermochromatographic experiments[3–5].

Existing CFD treatments of such wall exchange differ mainly in where the exchange is applied and in how many coupled species they carry. Lobresco et al.[6] compared a calibrated temperature-based deposition law with a vapor-pressure-driven Hertz–Knudsen–Schrage law for PbI_2, both formulated as volumetric source terms rather than boundary fluxes. For steam condensation with non-condensable gases, Vijaya Kumar et al.[7] implemented a diffusion-layer wall model in the OpenFOAM-based containmentFOAM both as volumetric sources and as face fluxes, and found the face-flux form more accurate, cheaper and less sensitive to the grid. Mixed (Robin) deposition boundary conditions for fission-product transport in OpenFOAM have been verified against analytical solutions[8]. In these studies each depositing or condensing species is transported on its own; gas-phase changes of chemical form are not coupled to the wall exchange.

This paper adds that coupling, in the finite-volume framework of OpenFOAM[9]. Our contribution has three parts. First, element-conserving homogeneous equilibrium among nine Pb–Bi–I gases is coupled to reversible face-flux wall exchange, in which the interfacial kinetic conductance acts in series with diffusion over the distance from the cell center to the wall; the Graetz–Robin verification shows that this face flux is second-order accurate where the same kinetic law applied to the cell-center value, the form of a first-cell sink, is first order. Second, gas pathways that lead to the same condensed material share one physical wall inventory, so that switching between deposition, evaporation and reaction cannot create or duplicate material. Third, we characterize the transport behavior within the tested model: evidence for predominantly diffusion-controlled wall exchange, profiles whose local shape is far more resolution-sensitive than their integrals, and an iodide partition that is controlled by thermodynamic input. The software also has an interface to the GEMS3K kernel of GEM-Selektor[10], but the reported calculations use a specialized ideal-gas element-potential solver.

All experimental input comes from published papers, so we report the consequences of two explicit source models and of selected input variations rather than infer a unique physical source. The study addresses three questions. Can transport, gas equilibrium and competing wall channels be coupled while preserving each represented element? How strongly do deposition profiles depend on time step, iteration, mesh and advection scheme, including when total deposition appears stable? Which input changes materially alter the interpretation of the computed deposits? The evidence comprises 79 computational cases of the three available Liu et al. configurations (not experimental replicates), production re-runs with a bounded advection scheme, a scheme and mesh study, an iteration check, the Graetz–Robin verification, and an untuned comparison with five single-species experiments of Lobresco et al. and with the two models of that study. The scope is the executed model: a fixed carrier and temperature field, prescribed release, ideal-gas speciation, and wall deposits without aerosol formation or substrate chemistry (section VIII C). Comparisons with experiment are conditional assessments of this model, not independent predictive validation.

Sections II to IV define the reconstructed inputs, equations and case matrix. Section V verifies the wall discretization and assesses conservation and resolution, and sections VI and VII examine experimental discrepancies and input sensitivity. Section VIII states what the evidence supports and collects the limitations.

TABLE I. Published configurations used in the baseline matrix. Sample masses are initial masses, not the amounts released in a 60 s run. Every experimental evaporation period was 10,800 s.

Identifier	Column	Diameter (mm)	Sample (mg)	He (mL/min)
PbI2_SS	Steel	4.8	14.6	100
LBE-I_SS_II	Steel	4.8	346.0	45
LBE-I_SiO2_I	Silica	5.0	368.3	45

II. EXPERIMENTAL BASIS AND INPUT RECONSTRUCTION

A. Available configurations

Liu et al.[1] used 1.25 m columns with an inner diameter of 4.8 mm for 316L stainless steel and 5.0 mm for fused silica. A 15 mm sample boat was inserted into a hot starting furnace, followed by a second furnace that imposed a decreasing temperature profile. Helium carried the evaporated material downstream. After three hours the source was withdrawn, the furnaces and gas flow were turned off, and the column cooled before the deposits were analysed. Temperature measurements and radiotracer scans used 1 cm increments.

Table I lists the three configurations for which both temperature curves and iodine scans could be reconstructed. The stand-alone BiI_3 experiment at 800 °C was not run, because its axial temperature profile is not published; it is neither given a surrogate profile nor counted as an unsuccessful simulation.

Both LBE samples had a reported iodine mole fraction of 9.04×10^{-4} and a 900 °C starting furnace. Their reported deposited iodine amounts were 0.191 and 0.203 mg, respectively. These amounts set the prescribed mean iodine source in the present cases; they are not measurements of a time-resolved evaporation rate or of inlet speciation.

B. Digitisation and coordinate assumptions

We digitized the axial temperature curves and iodine histograms from the published figures. Temperatures are interpolated linearly to cell centers and wall faces; beyond the plotted interval they are held at the nearest endpoint value. The computational coordinate is the plotted coordinate, whose origin relative to the physical boat and furnace is unconfirmed (fig. 1); the silica-coordinate study varies this assumption.

The recorded digitisation resolutions are approximately 1 mm in position, 4 K in temperature and 0.4 percentage points in histogram height. They describe raster extraction only, not coordinate registration, detector response, temperature reproducibility or calibration, and the uncertainties printed with the deposition temperatures are not uncertainties of every profile bin. The reconstructed histograms sum to 99.17%, 96.70% and 102.24% for pure PbI_2, steel LBE and silica LBE. The primary comparison keeps these sums and does not silently renormalise them.

In the LBE scans, the paper assigns 6.6% and 7.0% of the iodine to KI, which lies outside the Pb–Bi–I species set. We scale the model profile to the remaining 93.4% or 93.0%, while the measured histogram keeps all its published deposits. A second, explicitly conditional comparison normalises both profiles over their common scanned range (section VI). Neither comparison supplies the omitted potassium chemistry.

C. Simulation duration and measurement protocol

All source-driven cases run for 60 s with a source rate based on the 10,800 s experimental period, so they release 1/180 of the assumed three-hour inventory; the cumulative wall inventories inform deposition rates and conditional profile shapes but still contain the initial

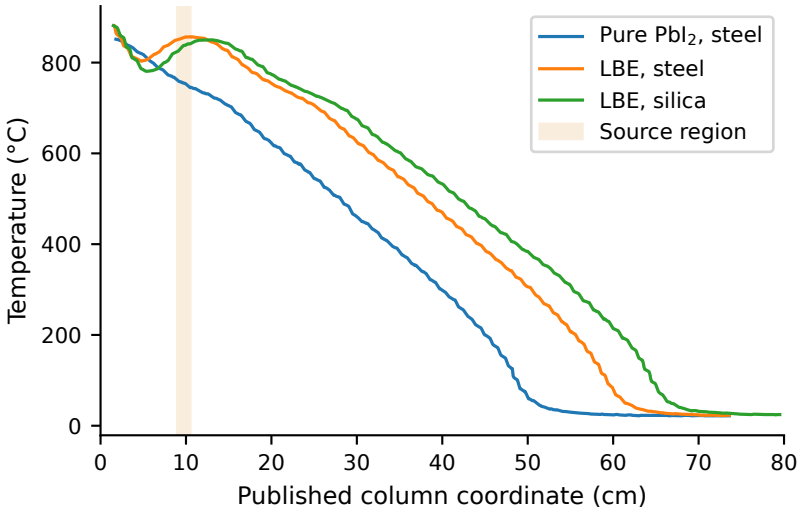


FIG. 1. Axial temperature curves digitized from Liu et al.[1] for the three configurations, with the assumed computational source interval. Beyond the plotted range, the CFD domain holds the endpoint values constant. The curves reconstruct published figures; they are not temperature fields predicted by a conjugate heat-transfer calculation.

filling transient. Four additional cases stop the source and the carrier flow at 60 s and continue to 80 s with the original temperature profile. They probe the relaxation of residual gas, not the experimental cooling history (supplementary material, Sec. S1).

III. TRANSPORT, SPECIATION AND WALL EXCHANGE

A. Frozen carrier and trace-species equations

The carrier fields ρ, $\mathbf{u}$, T and p come from a steady laminar helium calculation and are then held fixed. The reconstructed axial temperature is imposed over each whole cross-section. A gas species i is represented by Y_i, its mass per unit carrier mass, and satisfies

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\rho D_i \nabla Y_i) + S_i^{\mathrm{rel}} + S_i^{\mathrm{wall}}. \tag{1}$$

Gas equilibrium is applied as a local redistribution between transport steps. Species do not alter the carrier density, momentum or temperature. Here D_i is the molecular diffusivity and S_i^{rel} and S_i^{wall} are volumetric mass sources ($\mathrm{kg\,m^{-3}\,s^{-1}}$); the latter is positive for net wall release. Because Y_i is defined relative to carrier mass, the nine reactive-species variables are not constrained to sum to unity. The represented gases are Pb, PbI, PbI_2, Bi, Bi_2, BiI, BiI_3, I and I_2. Helium is inert. Condensed PbI_2, BiI_3, Pb and Bi are stored as immobile wall inventories. The solid-state identifiers in the code label these reservoirs; the equilibrium-pressure tables select between crystalline and liquid states where applicable.

The model does not represent bulk nucleation, aerosols and their thermophoretic deposition[11,12], thermal (Soret) diffusion[13,14], hydrogen/oxygen chemistry, latent-heat feedback or substrate reactions, so differences between the columns come from their thermal profiles, dimensions and sources, not from a substrate-specific sticking law.

B. Homogeneous ideal-gas equilibrium

Let ν_{ki} be the number of atoms of element $k \in \{\mathrm{Pb}, \mathrm{Bi}, \mathrm{I}\}$ in gas i. At fixed local temperature the transported element concentrations are

$$b_k = \sum_i \nu_{ki} \frac{\rho Y_i}{M_i}. \tag{2}$$

The equilibrium solver finds the gas partial pressures and dimensionless element potentials λ_k that satisfy

$$\begin{aligned} \frac{p_i}{p^\circ} &= K_{f,i}(T) \exp\left(\sum_k \nu_{ki} \lambda_k \right), \\ \sum_i \nu_{ki} \frac{p_i}{RT} &= b_k, \qquad p^\circ = 10^5\,\mathrm{Pa}. \end{aligned} \tag{3}$$

The tabulated $K_{f,i}$ share a common elemental reference and the stated standard pressure; the thermodynamic records, their provenance and the phase treatment are summarised in Sec. S5 of the supplementary material. Concentrations are converted back through $Y_i = M_i p_i/(\rho RT)$. Re-speciation therefore conserves elements, not individual species. The helium pressure is not an additional reactive element constraint; under the trace-species approximation its field stays prescribed.

The solver works in logarithmic variables with a safeguarded Newton iteration. The Pb and Bi subproblems are reduced at fixed iodine potential, and a Schur complement treats the remaining coupling. Re-speciation is requested at every time step in cells with $T \geq 350\,\mathrm{K}$; below this threshold the transported gas composition is kept. This cutoff is a numerical modeling choice, not a demonstration that reactions cease at 350 K. Homogeneous condensed phases are excluded, and instantaneous gas equilibrium stands in for unmeasured finite reaction rates.

C. Reversible iodide exchange

For PbI_2 and BiI_3, the kinetic wall law defines the flux as positive from condensate to gas:

$$j_i = G_i \left[p_{\mathrm{eq},i}(T_w) - p_{i,w} \right], \qquad G_i = C_e s_i \frac{2\alpha_i}{2-\alpha_i} \sqrt{\frac{M_i}{2\pi R T_w}}. \tag{4}$$

Here j_i is in $\mathrm{kg\,m^{-2}\,s^{-1}}$, M_i is in $\mathrm{kg\,mol^{-1}}$, pressures are in Pa, α_i is the accommodation coefficient and C_e, s_i are dimensionless multipliers. The baseline uses $C_e = s_i = \alpha_i = 1$. This is the Hertz–Knudsen–Schrage (HKS) form[15–17] used in vapor-pressure-driven deposition modeling[6]. The condensed-state pressure follows from the gas and condensed Gibbs energies on the stable tabulated condensed branch. The wall temperature selects this pressure; no deposition threshold is prescribed.

The kinetic law must also connect to the cell-center concentration through molecular diffusion. With d_f the cell-center-to-wall distance, $p_{i,c} = \chi_{i,c} Y_{i,c}$ and $\chi_{i,c} = \rho_c R T_c / M_i$, eliminating the wall concentration gives

$$\begin{aligned} G_{i,f}^{\mathrm{eff}} &= \frac{G_i}{1 + G_i \chi_{i,c} d_f / (\rho D_i)_f}, \\ j_{i,f} &= G_{i,f}^{\mathrm{eff}} [p_{\mathrm{eq},i}(T_w) - \chi_{i,c} Y_{i,c}]. \end{aligned} \tag{5}$$

This combines kinetic and half-cell diffusion resistances in series. In particular, $1/G_{i,f}^{\mathrm{eff}} = 1/G_i + \chi_{i,c} d_f/(\rho D_i)_f$: the smaller conductance limits transfer. As G_i increases, the effective conductance approaches the finite diffusive limit $(\rho D_i)_f/(\chi_{i,c} d_f)$. This separates the

physical interfacial coefficient from the mesh-dependent distance used to connect the wall to the resolved gas field. The gas equation receives the face flow $A_f j_{i,f}$ as an implicit cell contribution, and the wall inventory receives its negative. Evaporation is limited by the face inventory available over the time step, so a bare wall cannot evaporate. Inventory updates use the exchange actually realized by the final gas solve, including active bounds, not an independently evaluated explicit predictor. Switching between deposition and evaporation therefore does not create material.

D. Shared metal inventories and a reactive channel

Additional wall channels connect Pb(g) to condensed Pb, and Bi(g) and Bi_2(g) to one common condensed Bi inventory. Each channel has its own equilibrium pressure and stoichiometric mass conversion, and all use the baseline interfacial coefficients. The reactive channel

$$3\mathrm{BiI}(g) \longrightarrow 2\mathrm{Bi(condensed)} + \mathrm{BiI_3}(g) \tag{6}$$

is enabled only in the forward direction in the executed cases. For unit activity of condensed Bi, the dimensionless reaction constant and effective reactant equilibrium pressure are

$$K_r(T_w) = \exp\left[-\frac{2G^\circ_{\mathrm{Bi,c}} + G^\circ_{\mathrm{BiI_3,g}} - 3G^\circ_{\mathrm{BiI,g}}}{RT_w}\right], \tag{7}$$

$$p_{\mathrm{eq,BiI}} = p^\circ \left[\frac{p_{\mathrm{BiI_3,c}}/p^\circ}{K_r(T_w)}\right]^{1/3}, \tag{8}$$

$$j_{\mathrm{BiI}} = \min\{0,\, G^{\mathrm{eff}}_{\mathrm{BiI}}[p_{\mathrm{eq,BiI}} - p_{\mathrm{BiI,c}}]\}. \tag{9}$$

All Gibbs energies in K_r are evaluated at T_w. This phenomenological channel applies the HKS conductance, with the same half-cell resistance, to the reaction driving force; the coproduct pressure is taken from the beginning-of-step gas field, and the minimum suppresses the reverse reaction. The reaction returns one BiI_3 molecule to the gas and adds two Bi atoms to the shared wall inventory, on which all Bi channels draw; a separate inventory per reaction would duplicate a physical material. Coproducts are applied after the transported reactants have been solved. The channel is an irreversible model assumption, not a surface-equilibrium model.

E. Prescribed source models

Source model Q1 releases PbI_2 and, for LBE, BiI_3 at constant rates

$$q_{1,i} = \frac{m_i}{M_i \tau_{\mathrm{exp}}}, \qquad \tau_{\mathrm{exp}} = 10{,}800\,\mathrm{s}. \tag{10}$$

For pure PbI_2, $m_i = 14.6\,\mathrm{mg}$ is the initial sample mass, and complete uniform release over the experimental period is assumed. For steel LBE, $m_{\mathrm{PbI_2}} = 0.111\,\mathrm{mg}$ and $m_{\mathrm{BiI_3}} = 0.181\,\mathrm{mg}$; for silica LBE, they are 0.243 and 0.085 mg. These compound masses were inferred from the deposits reported in the experimental paper. Q1 therefore conditions both the iodine loading and much of the species split on the measurements.

Source model Q2 instead supplies atomic Pb, Bi and I to the gas-equilibrium system. With q_{He} the full-column helium molar flow and p_h the source pressure, its elemental rates are

$$q_{2,\mathrm{Pb}} = q_{1,\mathrm{PbI_2}} + q_{\mathrm{He}}\beta p_{\mathrm{Pb}}(T_h)/p_h, \tag{11}$$

$$q_{2,\mathrm{Bi}} = q_{1,\mathrm{BiI_3}} + q_{\mathrm{He}}\beta[p_{\mathrm{Bi}}(T_h) + 2p_{\mathrm{Bi_2}}(T_h)]/p_h, \tag{12}$$

$$q_{2,\mathrm{I}} = 2q_{1,\mathrm{PbI_2}} + 3q_{1,\mathrm{BiI_3}}. \tag{13}$$

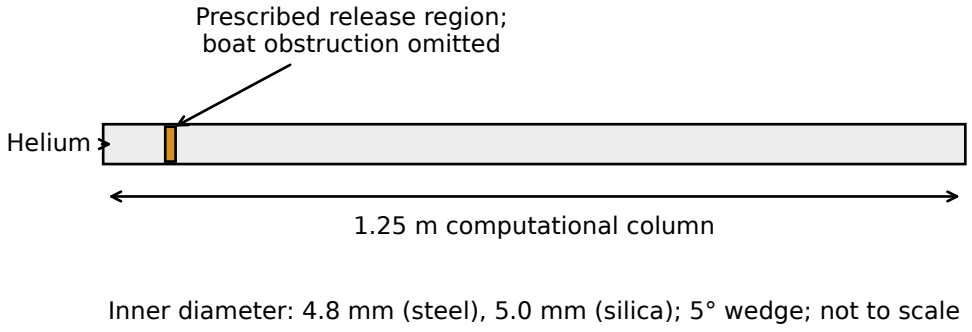


FIG. 2. Computational geometry and prescribed source region. The source is distributed over selected cells; the sketch does not show a resolved boat surface. The model keeps the tube length and the material-specific diameters, and the 5° wedge provides the axisymmetric reduction.

The metal pressures use ideal LBE activities $a_{\mathrm{Pb}} = 0.447$ and $a_{\mathrm{Bi}} = 0.553$ at $T_h = 1173.15\,\mathrm{K}$. The multiplicative metal saturation β is unmeasured. Q2 changes the metal supply and lets the iodide split emerge from the model, but its iodine source is still fixed by the deposited iodine. Q2 is therefore a source hypothesis, not an independent identification of LBE evaporation. The rates are distributed over an assumed source region, without resolving a liquid surface, depletion of the physical boat or the boat's obstruction of the flow.

IV. NUMERICAL IMPLEMENTATION AND STUDY DESIGN

A. Geometry, carrier and boundary conditions

A structured axisymmetric 5° wedge represents the 1.25 m column (fig. 2). The tube radius R_t is 2.4 mm for steel and 2.5 mm for silica. The production mesh has $576 \times 72 \times 1 = 41{,}472$ cells. The 64 follow-up sensitivities use the original $384 \times 48 \times 1 = 18{,}432$-cell mesh. The mesh studies use $256 \times 32 \times 1 = 8{,}192$, 18,432, 41,472 and, for one case, $864 \times 108 \times 1 = 93{,}312$ cells; successive levels refine the axial and radial spacing by a factor of 1.5. The source occupies $x = 0.09$–0.105 m and the radial interval $r = 0.25R_t$–$0.75R_t$. Its position and cell selection are computational assumptions, and the physical boat is omitted from the carrier geometry. Full-column source and inlet rates are multiplied by $5/360 = 1/72$ for the wedge.

The inlet flow rates are converted to helium mass flow assuming a controller reference of 298.15 K and 101,325 Pa, which the supplied paper does not specify. Full-column rates are $2.727 \times 10^{-7}\,\mathrm{kg\,s^{-1}}$ at 100 mL/min and $1.227 \times 10^{-7}\,\mathrm{kg\,s^{-1}}$ at 45 mL/min; a separate study uses a 273.15 K reference. Helium follows the perfect-gas law with constant heat capacity and Sutherland viscosity (supplementary material, Sec. S5). Steady laminar carriers are computed with `rhoSimpleFoam` while the reconstructed temperature profile stays prescribed. The converged face mass flux is corrected to discrete continuity within 10^{-12} of the inlet flow (supplementary material, Sec. S2), which makes the frozen transport field conservative but is not evidence that the velocity field has converged.

The boundary conditions are the same in all Liu et al. cases. At the inlet the carrier enters with the prescribed mass flow at the local wall temperature, and every reactive gas has a fixed mass fraction of zero: clean helium enters, and the release is a volumetric source in the source cells. At the outlet the static pressure is fixed at 101,325 Pa, an assumed value because neither experimental paper states the column pressure, and all gases have zero normal gradient, so they leave by advection. At the wall the transported gases have zero normal gradient in the transport operator, and all wall exchange enters as the face fluxes of section III C. With the 101,325 Pa outlet pressure and the baseline diffusivity,

the Schmidt number is 2.0–11.3 between 400 and 1173 K, and the diameter-based Péclet number $\mathrm{Pe} = \mathrm{Re}\,\mathrm{Sc}$ is 5.9–17.4 at 100 mL/min and 2.6–7.8 at 45 mL/min. Axial diffusion is therefore not negligible next to advection, and the fixed zero inlet value permits a diffusive loss of vapor through the inlet; it is included in the outward transport of the material ledger (eq. (15)). The flow-stop cases replace both end conditions by zero normal species flux (supplementary material, Sec. S1).

B. Transport regime

With the full-column helium mass flow $\dot{m}$, the tube diameter d and the Sutherland viscosity, $\mathrm{Re} = 4\dot{m}/(\pi d\mu)$ is 1.5–3.0 for the 100 mL/min steel case and 0.7–1.3 for the 45 mL/min cases between the source-furnace temperature of 1173 K and 400 K; the carrier is laminar throughout. With the baseline diffusivity, the radial diffusion time R_t^2/D_i is 58 ms in steel and 63 ms in silica. Placing the conductance of eq. (4) in series with diffusion over the tube radius, as in eq. (5) with d_f replaced by R_t and $T_c = T_w$, defines a wall Damköhler number

$$\begin{aligned} \mathrm{Da}_i &= \frac{G_i \chi_i R_t}{\rho D_i} \\ &= C_e s_i \frac{2\alpha_i}{2-\alpha_i} \sqrt{\frac{RT_w}{2\pi M_i}}\, \frac{R_t}{D_i}, \end{aligned} \tag{14}$$

where R is the universal gas constant and R_t is the tube radius. Because $\chi_i/\rho = RT_w/M_i$, it does not depend on pressure. For PbI_2 and BiI_3 between 400 and 1173 K, the baseline coefficients give $\mathrm{Da}_i \approx 1.4 \times 10^3$–$2.9 \times 10^3$, and the tested accommodation coefficients $\alpha_i = 0.1$ and 0.01 give about 76–150 and 7–15. This tube-scale estimate places the interfacial resistance well below the radial diffusion resistance for $\alpha_i = 1$ and 0.1, and about an order of magnitude below it for $\alpha_i = 0.01$. It is a scaling estimate, not a substitute for the resolved near-wall gradients of the calculations.

Two omitted effects can be bounded with the same inputs, the digitized wall profiles and the 101,325 Pa outlet pressure. The first is buoyancy. Neither experimental paper states the column orientation. For developed laminar flow with a linearly varying wall temperature, the wall-to-bulk temperature difference is $(11/192)\,U d^2 |\mathrm{d}T_w/\mathrm{d}x|/a$, with a the thermal diffusivity of helium; it stays below 2.6 K. A Grashof number based on the axial temperature change over one diameter, $\mathrm{Gr} = g d^4 |\mathrm{d}T_w/\mathrm{d}x|/(T\nu^2)$, stays below 2.2, and below 1.4 where the wall temperature lies between 400 and 750 K. Because Re is also of order one, the Richardson number $\mathrm{Gr}/\mathrm{Re}^2$ reaches 0.15 (pure PbI_2) to 0.78 (silica) in that range, and about 1.2 in the steepest cold section of the silica profile. Compositional buoyancy at the largest computed trace mole fraction, 9.2×10^{-5} evaluated with the molar mass of BiI_3, gives $\mathrm{Gr}/\mathrm{Re}^2 \leq 0.32$. Richardson numbers of order 0.1–1 do not show that buoyancy is negligible: in a horizontal column, buoyancy-driven secondary flow may be significant locally, in particular in the steep cold sections where deposition occurs. The frozen carrier does not represent it, and with the orientation unknown its effect on the deposition profiles is an unquantified limitation of the model. The second effect is thermal (Soret) diffusion. Taking a thermal diffusion factor $\alpha_T = 1$ as a generous bound, the Soret drift velocity $D_i \alpha_T |\mathrm{d}T_w/\mathrm{d}x|/T$ is at most 2.2% of the carrier velocity. The radial Soret flux, relative to a Fickian flux $x_i D_i/R_t$ towards the wall, is at most $\alpha_T \Delta T_r/T \leq 0.7\%$, with ΔT_r the wall-to-bulk difference above. Both are small compared with the factor-of-two diffusivity variations examined in section VII.

C. Discretisation and coupling

The transient species equations use implicit Euler time integration and corrected Gauss linear diffusion. The production cases (the 15 baselines and four silica temperature-coordinate cases) use the bounded OpenFOAM scheme `Gauss limitedLinear 1`[18,19] for

advection on 41,472 cells. It blends linear and upwind interpolation through a TVD-type flux limiter, which reverts to upwinding near extrema and steep gradients. The 64 follow-up cases and the time-step comparisons keep the first-order upwind advection of the original campaign on 18,432 cells, and the follow-up sensitivities are always reported as differences from their own upwind baselines on that mesh. The scheme and mesh study adds the van Leer limiter[20] (`Gauss vanLeer`) as a second bounded scheme. All nine transported gases use the same advection scheme within a case. The baseline time step is 2 ms. Wall exchange is assembled implicitly with nonnegative source and sink coefficients and shared inventory bounds. Configured species correctors and active-bound guards reconcile the final gas solution with the realized wall transfer. Gas re-speciation then redistributes species under the local element constraints. The sequence is first order in time; a conservative equilibrium projection alone does not remove splitting or transport error. The baseline configuration requests three species correctors with a 10^{-16} tolerance and up to eight bound guards with a 10^{-13} guard tolerance; the effect of this stopping criterion is tested in section V B.

Every gas has the baseline diffusivity $D_i = 10^{-4}\,\mathrm{m^2\,s^{-1}}$. Three transport alternatives halve or double this value for all gases, or replace only the PbI_2 coefficient with a temperature- and pressure-dependent PbI_2–He estimate whose collision parameters are themselves estimates (supplementary material, Sec. S5). Accommodation variations set $\alpha = 0.1$ or 0.01 for the two reversible iodide pairs and keep the baseline metal-channel coefficients.

D. Completed case matrix

The 15 baseline cases are Q1 for each available configuration, plus Q2 for the two LBE columns at $\beta \in \{0.001, 0.01, 0.1\}$ with two thermodynamic branches. The second branch adds a constant 29 kJ/mol to the standard Gibbs energy of BiI(g); it is a sensitivity surrogate, not a second dataset. The 64 follow-up cases vary mesh and time step, diffusivity, accommodation, source position, Q2 metal loading, flow reference, silica temperature coordinate and a fixed-temperature flow stop; they are a structured set of conditional comparisons, not a factorial uncertainty study. Four further sets complete the evidence: the 19 production re-runs (section V), a scheme and mesh study of two cases up to 93,312 cells, three full-run corrector checks, and ten runs of five single-species experiments of Lobresco et al.[6] on two meshes (section VI C). All Liu et al. calculations and the Lobresco runs use the same solver executable, identified by its SHA-256 in the saved reports. Table S1 and Sec. S1 of the supplementary material list every case, its variation and its reference case.

V. NUMERICAL VERIFICATION AND RESOLUTION LIMITS

This section verifies the wall-exchange discretization on a problem with a reference solution, then reports material conservation and the time-step, iteration, mesh and scheme dependence of the iodine calculations.

A. Verification of the wall-exchange discretization

The half-cell series law of eq. (5) is verified on an extended Graetz problem with a Robin wall condition[2]: fully developed laminar flow in a pipe, axial diffusion included, and a constant first-order wall uptake $-\rho D\,\partial Y/\partial r = \rho k_w Y$ at $r = R$. Far from the inlet and the outlet every cross-sectional measure decays as $\exp(-\mu x)$, and the decay rate depends only on the Péclet number $\mathrm{Pe} = U_b R/D$ and the Biot number $\mathrm{Bi} = k_w R/D$, the analogue of the wall Damköhler number of eq. (14). The reference μR for $\mathrm{Pe} = 5$ and $\mathrm{Bi} = 1$ and 50 is the smallest positive eigenvalue of the radial problem, computed by a power series

TABLE II. Graetz–Robin verification (Pe = 5): relative error $\mu/\mu_{\mathrm{ref}} - 1$ of the fitted decay rate on N_r radial cells, observed order p and $\mathrm{GCI}_{\mathrm{fine}}$[21] on $N_r = 64/128/256$, Gauss linear advection, zero-gradient outlet. Half-cell series: eq. (5); cell-center value: kinetic law without the half-cell resistance.

Bi	Wall law	$N_r = 32$	64	128	256	p	GCI (%)
1	Half-cell series	9.7×10^{-5}	2.4×10^{-5}	6.1×10^{-6}	1.5×10^{-6}	2.001	0.000189
1	Cell-centre value	1.0×10^{-2}	5.0×10^{-3}	2.5×10^{-3}	1.3×10^{-3}	1.008	0.155
50	Half-cell series	-5.0×10^{-5}	-1.3×10^{-5}	-3.2×10^{-6}	-7.9×10^{-7}	1.992	0.0001
50	Cell-centre value	2.2×10^{-2}	1.1×10^{-2}	5.5×10^{-3}	2.7×10^{-3}	1.004	0.341

in extended precision, by shooting and by Chebyshev collocation; the three agree within a relative 1.69×10^{-12}.

The test uses the solver's own wall-exchange implementation with a constant uptake velocity k_w ($R = 2.4$ mm, $T = 500$ K, $p = 1$ atm, $D = 7.6963 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$, a pipe of length $24R$). The meshes are 2° wedges with exactly axisymmetric cell metrics and $N_r = 32$, 64, 128 and 256 uniform radial cells, $\Delta x = 2\Delta r$, and the parabolic carrier enters through exact face fluxes. Steady states are obtained by implicit Euler steps with a very large time step, and μ is fitted to the computed gas line density in a window chosen from the spectrum. Two wall laws are compared: the half-cell series law and the kinetic law applied directly to the cell-center value (no half-cell resistance), which is the form of a volumetric first-cell sink. Every fitted μ equals the decay rate of the developed discrete mode of the documented scheme within 7.09×10^{-9}, and moving the fit window or changing the outlet condition changes it by less than 10^{-7}, so the differences below are discretization errors. All 27 acceptance criteria of the study pass (records, provenance and regeneration: Sec. S7 of the supplementary material).

Table II gives the relative error of μ on each mesh, the observed order and the grid convergence index $\mathrm{GCI}_{\mathrm{fine}}$ of Celik et al.[21] on $N_r = 64/128/256$. The half-cell law converges at order 1.99–2.00 with $\mathrm{GCI}_{\mathrm{fine}} \le 1.89 \times 10^{-4}\%$, and its Richardson-extrapolated decay rate matches the reference within 9.47×10^{-9}. The cell-center law converges at first order (1.004–1.008) with $\mathrm{GCI}_{\mathrm{fine}} = 0.155$–$0.341\%$. With bounded SuperBee advection the observed orders are 2.00–2.01 and 1.01–1.02, respectively (supplementary material, Table S4). On the coarsest mesh the cell-center law errs by 1–2% in the decay rate, the half-cell law by at most 10^{-4}.

This test verifies the discretization of the wall exchange and its coupling to advection and diffusion for a linear wall law at $\mathrm{Bi} \le 50$. It does not verify the nonlinear HKS law with tabulated equilibrium pressures, the gas equilibrium or the wall inventories, and it does not show that the full iodine calculations are mesh-converged; their resolution is assessed separately below. The iodide Damköhler numbers of section IV B also exceed the tested Biot numbers, and the meshes of the iodine calculations have cell aspect ratios far from the $\Delta x = 2\Delta r$ of the test.

B. Material balance and solver diagnostics

All 15 runnable baseline cases, all 64 follow-up cases, the 19 production cases and the three full-run corrector cases completed and passed their configured numerical acceptance checks; the scheme and mesh runs completed with the same ledgers. This PASS classification is not an experimental-validation classification. For element k the raw material residue is

$$R_k = N_k^{\mathrm{gas}} + N_k^{\mathrm{wall}} + N_k^{\mathrm{sample}} + N_k^{\mathrm{net\,out}} + N_k^{\mathrm{removed}} - N_k^{\mathrm{initial}} - N_k^{\mathrm{released}}, \tag{15}$$

where N_k are molar atom inventories and outward transport is signed; the linear-solve defect and clamping corrections are recorded separately and are not inserted into R_k. Divided by

TABLE III. Iodine profile changes E_{L1} (percent) between successive mesh levels at 2 ms, relative to the finer level, for three advection schemes, and the effect of halving the time step on 18,432 cells with upwind advection. All steel cases are Q2; the shifted-BiI case uses 1% metal saturation. A dash marks a combination that was not run.

Case	Scheme	8,192→18,432	18,432→41,472	41,472→93,312	2→1 ms
Pure PbI_2	upwind	6.57	5.06	–	0.0043
	limitedLinear	5.70	2.15	–	–
	van Leer	4.94	2.23	–	–
Steel, shifted BiI	upwind	10.47	10.58	2.13	0.1536
	limitedLinear	15.35	8.95	1.98	–
	van Leer	16.83	9.17	–	–
Steel Q2, 0.1%	upwind	9.02	5.26	–	0.2326
Steel Q2, 1%	upwind	9.72	2.06	–	0.3228

the supplied material and maximised over steps and elements, the largest raw residue is 1.41×10^{-12} for the 19 production cases and 8.85×10^{-13} for the 60 source-driven follow-up cases. The calculations conserve the represented elements to round-off; this says nothing about physical accuracy or discretization error. The ledger terms, the equilibrium-solver diagnostics, the startup sign flags and the trace-loading bound are given in Sec. S2 of the supplementary material.

Some paired wall-coupling iterations do not meet the requested 10^{-16} outer criterion. Three cases were therefore repeated over the full 60 s with eight instead of three correctors (18,432 cells, upwind). The iodine profiles of pure PbI_2 and of the steel Q2 case at 0.1% saturation change by 1.09×10^{-13}% and 4.89×10^{-13}%, that is, at round-off; the steel case at 1% saturation changes by 0.021%. The requested tolerance lies below the attainable residual floor (Sec. S2), and the iteration error of the reported profiles is negligible next to the time-step and spatial errors below.

C. Time-step, mesh and scheme dependence

For profiles on the common 1 cm bins, let a_b be the deposited iodine atoms in bin b for a comparison case and a_b^0 the corresponding reference inventory. We report the profile change E_{L1} and the total change ΔN_I:

$$E_{L1} = 100 \frac{\sum_b |a_b - a_b^0|}{\sum_b a_b^0}, \qquad \Delta N_I = 100 \frac{\sum_b a_b - \sum_b a_b^0}{\sum_b a_b^0}. \tag{16}$$

Both measures retain changes in amount. Source, physical coefficients, equilibrium engine and executable are fixed within each numerical comparison.

Table III separates time-step from mesh and scheme effects; for successive mesh levels the reference a_b^0 is the finer level. Reducing the time step from 2 to 1 ms on 18,432 cells changes the iodine profile by 0.0043–0.3228%. With the upwind scheme of the original campaign, the 18,432 to 41,472-cell differences of the four follow-up cases are 2.06–10.58%, while total deposited iodine changes by less than 0.014% and the centroid moves by at most 0.0652 cm. Integral quantities are thus stable while local profile shapes still differ appreciably.

The scheme and mesh study (table III and fig. 3) addresses the shifted-BiI steel case, which had the largest upwind change. With limited-linear advection its iodine profile changes by 15.35, 8.95 and 1.98% over the three refinements to 93,312 cells; with upwind, by 10.47, 10.58 and 2.13%. Pure PbI_2 changes by 5.70 and 2.15% with limited-linear, against 6.57 and 5.06% with upwind. The van Leer scheme differs from limited-linear by at most 2.8% (steel) and 1.8% (pure PbI_2) at equal mesh. Between 41,472 and 93,312 cells, the 3 cm smoothed profile of the steel case changes by 0.67% and its total deposit by 0.015%; over all

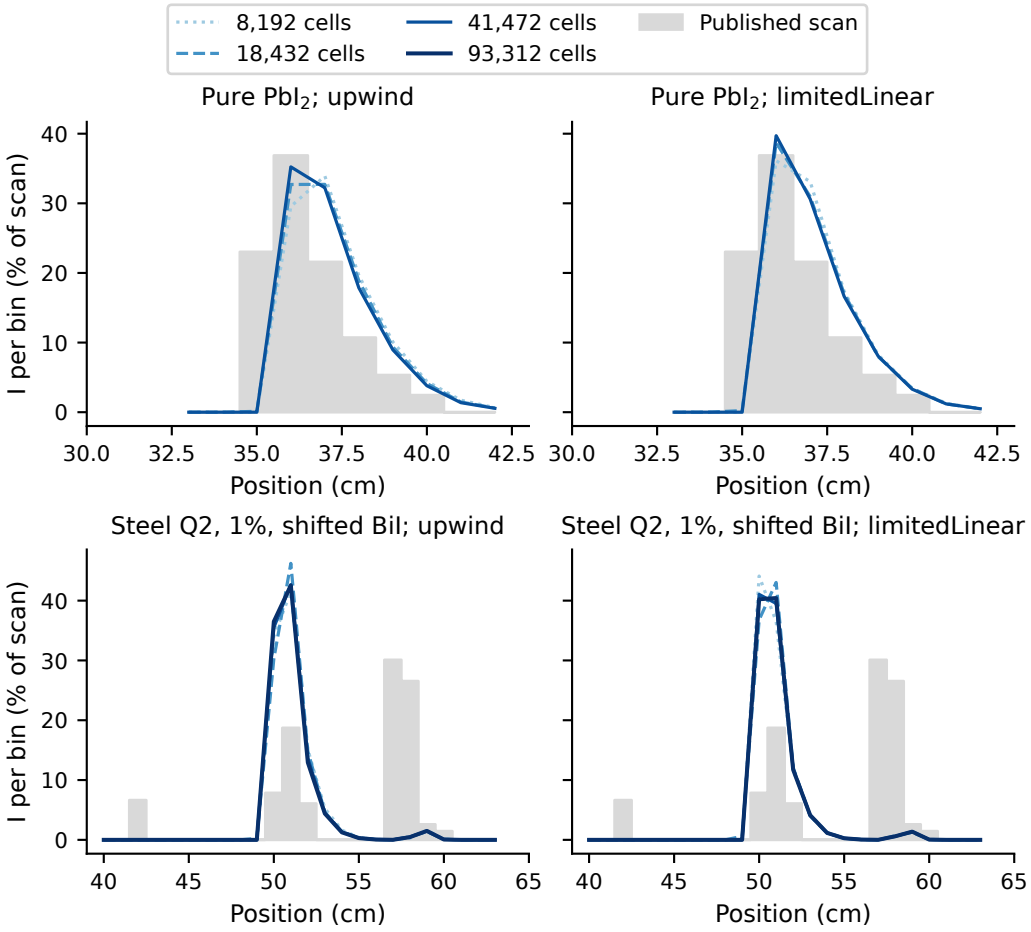


FIG. 3. Mesh sequences of the iodine deposition profile for pure PbI_2 (top) and the shifted-BiI steel case at 1% metal saturation (bottom), with first-order upwind (left) and limited-linear (right) advection. Each curve is normalized over the common scanned interval so that shapes can be compared; the unnormalised differences are given in table III. Grey histograms are the reconstructed experimental iodine scans[1] and include any KI contribution.

four levels its centroid stays within 0.6 mm and the wall temperature at the PbI_2 centroid within 0.9 K.

The successive limited-linear profile changes of the steel case give apparent orders of 1.3 and 3.7, and its iodine total, centroid and width oscillate on both triplets. These orders are inconsistent, so no formal grid convergence index is claimed[21–23]. We quote the measured 41,472 to 93,312-cell change of 1.98% as the spatial uncertainty indicator of the production profiles. Extrapolating the last change at the last apparent order would place the 41,472-cell profile about 2.5% from the mesh limit, and assuming second order about 3.6%. These are indicative estimates for one case, not confirmed bounds, and they do not establish mesh independence of local profile shapes in other cases. The three-grid scalar checks of the earlier upwind triplets are in Sec. S3 of the supplementary material.

At equal mesh, upwind differs from limited-linear by 20.8, 14.1, 11.3 and 7.7% for the steel case on the four levels and by 9.0–13.2% for pure PbI_2, much more than the two bounded schemes differ from each other. This is consistent with the local estimate $D_{\mathrm{num}} \simeq |u_x|\Delta x/2$, which is comparable to or greater than the prescribed $10^{-4}\,\mathrm{m^2\,s^{-1}}$ in parts of the carriers. Upwind numerical diffusion is therefore the dominant spatial error of the original campaign, and it is why halving the time step could not establish convergence of profile widths. Across

the 19 production cases, the production profiles differ from the 18,432-cell upwind profiles by 9.1–22.3%. The follow-up sensitivities of section VII remain upwind differences on 18,432 cells, taken against baselines with the same numerical error.

D. Binned maxima and observation width

A deposition maximum picked on a 1 cm grid is fragile when adjacent bins have similar heights. In pure PbI_2 the 18,432-cell upwind maximum is at 37 cm and 350.7 °C, while the 36 cm bin is only about 0.05% lower; on 41,472 cells it moves to 36 cm and 366.5 °C although the centroid shifts by only −0.0585 cm. In the limited-linear steel sequence the PbI_2 maximum alternates between two nearly equal bins, 50 and 51 cm (306.3, 288.1, 306.7 and 288.1 °C on the four levels), while the 3 cm smoothed maximum stays at 51 cm and 288.0–288.1 °C and the centroid temperature at 290.6–291.5 °C. A one-bin switch can change the reported peak temperature by tens of kelvin without a comparable displacement of the profile, so table IV also reports the smoothed and centroid values. Conversely, comparing profiles on wider bins does not remove the mesh sensitivity: for the shifted-BiI case the original upwind fine/medium difference on 4 cm bins ranges from 0.52% to 10.02% depending on the bin origin (Sec. S4 of the supplementary material, which also gives a transport distance between the binned profiles).

VI. CONDITIONAL COMPARISON WITH DEPOSITION MEASUREMENTS

This section compares the calculations, mainly the original Q1 cases, with the reconstructed deposition scans of Liu et al., and the single-species model with the experiments of Lobresco et al. The comparisons are conditional on the reconstructed inputs and assumptions of sections II and III.

A. Comparison measures

Against experiment we use reconstructed relative iodine yields per 1 cm bin. Let e_b be the experimental percentage and let

$$m_b = (100 - y_{\mathrm{KI}}) \frac{a_b}{\sum_{\text{full column}} a_b} \tag{17}$$

be the modelled percentage, where the stated excluded KI yield is $y_{\mathrm{KI}} = 0$, 6.6 or 7.0. Over the common scanned bins $\mathcal{B}$, we report the raw discrepancy

$$L_{\mathrm{raw}} = \sum_{b \in \mathcal{B}} |m_b - e_b| \quad \text{(percentage points)}, \tag{18}$$

which keeps the experimental digitisation sum and the model's explicit KI exclusion. A secondary shape diagnostic normalises both profiles over $\mathcal{B}$, $\widehat{m}_b = m_b / \sum_{\mathcal{B}} m_b$ and $\widehat{e}_b = e_b / \sum_{\mathcal{B}} e_b$, and computes the overlap

$$O = 100 \sum_{b \in \mathcal{B}} \min(\widehat{m}_b, \widehat{e}_b) = 100 \Big[1 - \tfrac{1}{2} \sum_{b \in \mathcal{B}} |\widehat{m}_b - \widehat{e}_b| \Big]. \tag{19}$$

Identical binned shapes give 100% and disjoint shapes zero. O discards differences in capture and total yield, and its experimental normalisation still contains KI. L_{raw}, O and the simulation-to-simulation change E_{L1} (eq. (16)) have different denominators and are not interchangeable percentages of model accuracy.

TABLE IV. Q1 iodide maxima on the production mesh (41,472 cells, limited-linear advection), compared with the published peaks. Subscript bin: 1 cm binned maximum; subscript 3: maximum of the 3 cm smoothed profile; T_c: wall temperature at the deposit centroid. Positions are in cm and temperatures in °C. Experimental temperature ranges are those printed by Liu et al.[1]; they are not statistical confidence intervals for model error.

Case	Species	x_{exp}	x_{bin}	x_3	T_{exp}	T_{bin}	T_3	T_c
Pure PbI_2, steel	PbI_2	36	36	37	368±18	366.5	350.5	348.7
LBE, steel	PbI_2	51	51	52	288±18	288.1	269.2	274.5
LBE, steel	BiI_3	57	57	58	154±14	166.9	138.8	158.1
LBE, silica	PbI_2	51	56	56	306±24	287.8	287.8	288.8
LBE, silica	BiI_3	59	63	64	127±39	152.8	119.8	147.1

B. Q1 peaks and complete profiles

Table IV compares the five available Q1 iodide maxima with the printed peak temperatures. Besides the 1 cm binned maximum it lists the maximum of the 3 cm smoothed profile (x_3, T_3) and the wall temperature at the deposit centroid (T_c). All five binned peak temperatures lie within the printed ranges, but the other two measures do not always. For pure PbI_2 the binned maximum (36 cm, 366.5 °C) matches the published 36 cm and 368 °C; the smoothed maximum (350.5 °C) lies at the lower limit of the printed 368 ± 18 °C, and the centroid temperature (348.7 °C) lies 1.3 K below that 350 °C limit. For steel PbI_2 the binned value is 288.1 °C, the centroid temperature (274.5 °C) lies near the lower limit of 288 ± 18 °C, and the smoothed maximum (269.2 °C) lies 0.8 K below its 270 °C limit. The steel BiI_3 smoothed maximum (138.8 °C) falls just below its range, while the centroid temperature (158.1 °C) lies inside it. Whether a computed peak temperature agrees with a printed range therefore depends on which measure of the peak is used (section V). Steel LBE places both iodide maxima at the measured positions, whereas silica LBE places the PbI_2 and BiI_3 maxima 5 and 4 cm downstream; agreement with a printed temperature does not establish the spatial location.

Figure 4 shows the complete profiles. The raw discrepancies L_{raw} are 46.33, 52.97 and 183.84 percentage points for pure PbI_2, steel LBE and silica LBE (53.90, 53.45 and 181.84 on the 18,432-cell upwind mesh), and the corresponding normalized overlaps are 76.97%, 72.13% and 6.13% (72.93%, 71.88% and 7.20%). The silica case has a large profile discrepancy even though both printed temperature ranges contain its calculated binned maxima.

The Q1 steel BiI_3 share of deposited iodide iodine is 65.34%, close to the measured iodide-only share of $61.3/(61.3 + 32.1) = 65.63\%$; for silica the values are 28.68% and $27/(27 + 66) = 29.03\%$. Because the Q1 source masses were inferred from the same deposits, these agreements cannot serve as independent tests of gas speciation or LBE release. Similarly, the source-driven calculations capture 91.28–98.20% of the supplied iodine on the wall by 60 s, which says nothing about the experimental three-hour capture.

C. Single-species comparison: Lobresco et al.

The five PbI_2 thermosublimatography experiments of Lobresco et al.[6] vary the helium flow (5, 106 and 540 mL/min) and the duration (6, 60 and 300 s) with 1.37–2.23 mg samples in a 4.8 mm steel column. Unlike the Liu et al. cases, they do not require a deposit-derived source. That study also reports the deposition temperatures of two of its own CFD models: a temperature-based model calibrated against one reference experiment, and a vapor-pressure-based Hertz–Knudsen–Schrage model with a near-wall interaction layer and an effective rate multiplier C_e. We ran the five cases without adjusting any parameter to these data.

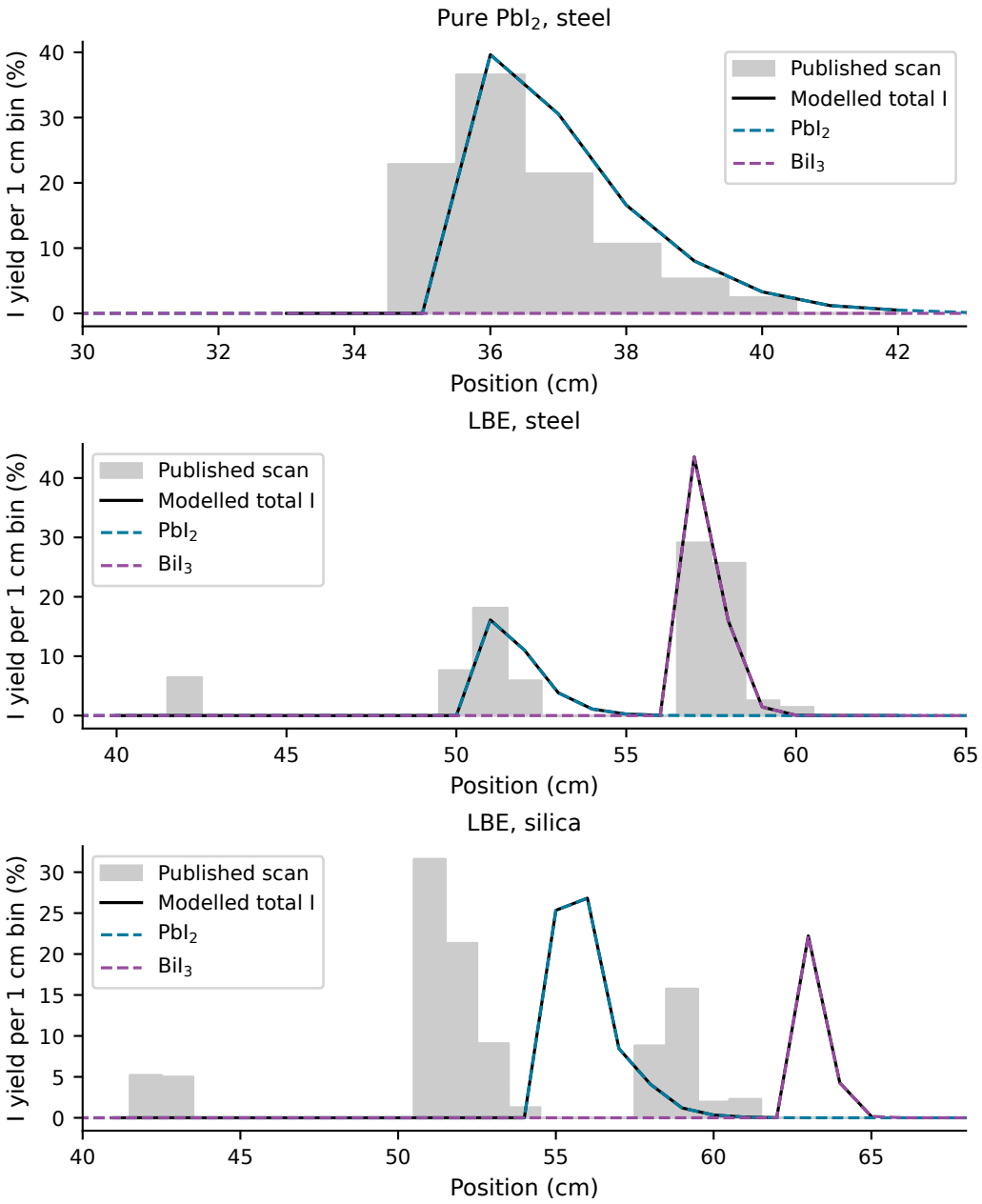


FIG. 4. Production Q1 iodine profiles (41,472 cells, limited-linear advection) compared with the digitized published scans (grey histograms)[1]. Solid black lines are total modelled wall iodine; dashed lines show its two iodide contributions. Model totals are normalized to 100%, 93.4% and 93.0% for pure PbI_2, steel LBE and silica LBE, respectively, to expose the omitted KI yield; experimental bars keep their digitized values.

a. Comparison region. The deposition temperature is defined as in that study: the wall temperature at the first position where the deposit exceeds 1% of its peak, linearly interpolated between 1 cm points. We apply this rule to the measured profiles from their first digitized point, and to our computed deposits downstream of the end of the sample, in all five cases. We take the rule to locate the onset of the main deposit formed by transport down the gradient, and the measured profiles contain only that deposit: the measured

T5_60 profile, digitized from 2 cm, has no deposit above the threshold upstream of 37 cm. For four cases the restriction makes no difference. At 5 mL/min, however, computed vapor diffuses upstream against the slow carrier through the zero-total-flux inlet condition (below), and 1.2% of the T5_60 deposit forms upstream of the sample, near the cold inlet. Applied from the inlet, the rule picks this deposit at 548 K instead of the main-deposit onset at 836 K, and the RMS deviation of our five onsets becomes 135 K instead of 18 K. The T5_60 value, and with it the RMS, therefore depends on the choice of comparison region; the computed upstream deposit is itself a discrepancy with the measurement.

b. Inputs. Flows (standard volumetric rates at 273.15 K and 101,325 Pa), durations and sample amounts are those of Table 1 of that study. The wall temperature of each flow rate, the measured profiles, the sample position and both published model curves were digitized from its Figs. 13–14 (CC BY 4.0); the 1% rule applied to the digitized measured profiles reproduces the published experimental deposition temperatures within 1.0 K. The model is that of sections III and IV reduced to one species: an expanding laminar helium carrier computed with the wall temperature, HKS exchange with $C_e = \alpha = 1$ in series with the half-cell diffusion resistance, the PbI_2–He diffusivity correlation of that study, our own phase-resolved PbI_2 vapor-pressure table, an inlet condition of zero total species flux, and an evaporating sample of the Table 1 amount at the digitized position, removed at the end of the experiment. Mesh spacings follow the production and 18,432-cell meshes (22,896 and 10,176 cells), with limited-linear advection; details, including a wall-temperature table of the T-Flows test case that differs from the published curves, are in Sec. S6 of the supplementary material. Our calculation thus differs from both published models in more than the wall treatment: they prescribe a fully developed laminar velocity profile, take the PbI_2 vapor pressure from a gas–solid equilibrium calculation, and represent the source as a timed inlet release (temperature-based model) or a volumetric solid inventory (vapor-pressure model).

c. Results. Table V and fig. 5 summarise the comparison. Our onset temperatures differ from the measurements by +26, +19, +16, −11 and −14 K, an RMS deviation of 18 K, against 35 K for the calibrated temperature-based model and 35 K for the vapor-pressure-based model. The two meshes agree within 0.2 K. The measured decrease of the onset temperature with duration at 106 mL/min is reproduced with a warm offset of 16–26 K; the onsets at 5 and 540 mL/min are 11–14 K too cold. Three of the five deviations exceed the stated ±15 K experimental temperature uncertainty. The profiles agree less well than the onsets: the computed peaks lie 1–4 cm upstream of the measured peaks, and the summed absolute profile difference is 0.37–0.89 of the sample amount (table V).

d. Limits of the comparison. Three limits restrict what the onset agreement can show. First, the comparison-region choice above decides the T5_60 result. Second, the near-source state lies outside the model: next to the sample the computed PbI_2 mole fraction reaches 0.33–0.36, which violates the dilute, frozen-carrier assumption (the vapor would alter density, velocity and diffusion), and the boat is not represented. Third, the release history is not constrained. In the model the sample is fully evaporated before the end of even the shortest, 6 s experiment: by then 98.1% of the T106_6 sample is deposited and the rest is gaseous. Lobresco et al. report that complete evaporation was confirmed in the experiments and that the evidence indicates near-complete evaporation within the shortest experiment, which supports release within 6 s. No measurement resolves the release on shorter time scales; the heat-up of the inserted boat and the actual sample surface are unknown, and the computed release rate comes from the non-dilute region where the model does not apply. The experiments also end by stopping the flow, withdrawing the sample and pulling the column out of the furnace, which is not simulated; counting the vapor still in the tube at the end time changes only T5_60 noticeably, to +3 K, because 4.0% of its inventory is still gaseous (RMS 17 K).

The onset comparison is therefore promising but conditional. Carrier, vapor-pressure and source assumptions differ from those of the published models, so the smaller RMS deviation cannot yet be attributed to the wall treatment, and the profile discrepancies above remain. It is an untuned comparison with published measurements and published model results, not a validation.

TABLE V. Deposition-onset temperatures for the five PbI_2 experiments of Lobresco et al.[6]: flow $\dot{V}$ (mL/min, standard), duration t (s), measured onset T_{exp} (K, their Table 2), and the onset of this work and of their temperature-based and vapor-pressure-based models minus T_{exp} (K; the latter two from their Table 2). Δx_{pk} is our deposit peak minus the measured peak (cm), and L_1/n_0 the summed absolute difference of our deposit and the digitized measured profile divided by the sample amount. This work: 22,896 cells, deposit at the end of the experiment, onset rule applied downstream of the sample in every case.

Case	$\dot{V}$	t	T_{exp}	This work	Temp.-based	Vap.-press.	Δx_{pk}	L_1/n_0
T106_6	106	6	814	+26	−39	+27	−4	0.89
T106_60	106	60	723	+19	−10	+37	−2	0.66
T106_300	106	300	667	+16	−1	+34	−2	0.88
T5_60	5	60	847	−11	−60	−50	−1	0.37
T540_60	540	60	700	−14	−30	−15	−1	0.47
RMS				18	35	35		

VII. INPUT SENSITIVITY AND INTERPRETATION

This section examines how source loading, BiI thermodynamic data, the silica temperature coordinate, and transport, kinetic and flow inputs change the computed deposits.

A. Source loading and thermodynamic input

The Q2 cases expose a sensitivity that is hidden when the iodide split is prescribed. Figure 6 shows the BiI_3 share of deposited iodide iodine,

$$f_{BiI_3} = \frac{3N^{wall}_{BiI_3}}{2N^{wall}_{PbI_2} + 3N^{wall}_{BiI_3}}, \tag{20}$$

which counts only the iodine in the two modelled iodides and therefore differs from the BiI_3 fraction of all experimental iodine, which also contains KI.

At 1% metal saturation on the production steel mesh, the baseline thermodynamics give $f_{BiI_3} = 94.58\%$ (94.57% with upwind advection). Raising the BiI(g) standard Gibbs energy by 29 kJ/mol, consistently in the formation and reactive-wall constants, gives 2.01% (2.00%). A constant shift of this size scales the formation constant by $\exp[-29{,}000/(RT)]$, approximately 0.051 at 1173 K and 0.00093 at 500 K. An independent integration of the saved species profiles of the original upwind campaign gives 94.532–94.569% and 1.922–2.004% over three meshes and the half-step run, a separation of 92.555–92.619 percentage points at every matched resolution. The response is therefore not numerical noise. What it measures is the importance of the BiI thermodynamic input for the predicted iodide partition: a single, arbitrarily sized perturbation moves the partition almost from one iodide to the other. The shift is not an estimate of the uncertainty of the BiI data, the two branches do not bound an uncertainty interval, and the comparison with the measured split is not a basis for selecting either dataset.

Source loading also matters. With baseline thermodynamics, the steel BiI_3 share at 0.2%, 0.316% and 0.562% metal saturation (follow-up cases, 18,432 cells, upwind) is 52.50%, 70.58% and 94.56%; the shifted branch gives 47.66%, 37.43% and 15.88%. The low-loading trend is not universally monotonic. The steel 0.1% case with baseline thermodynamics gives 56.48% and an 80.66% scan overlap (56.47% and 81.10% on the 18,432-cell upwind mesh), a useful diagnostic comparison but not an identified physical saturation. The published deposits alone cannot separate source loading, thermodynamic data and omitted chemistry; choosing a branch because it best matches one scan would be a calibration.

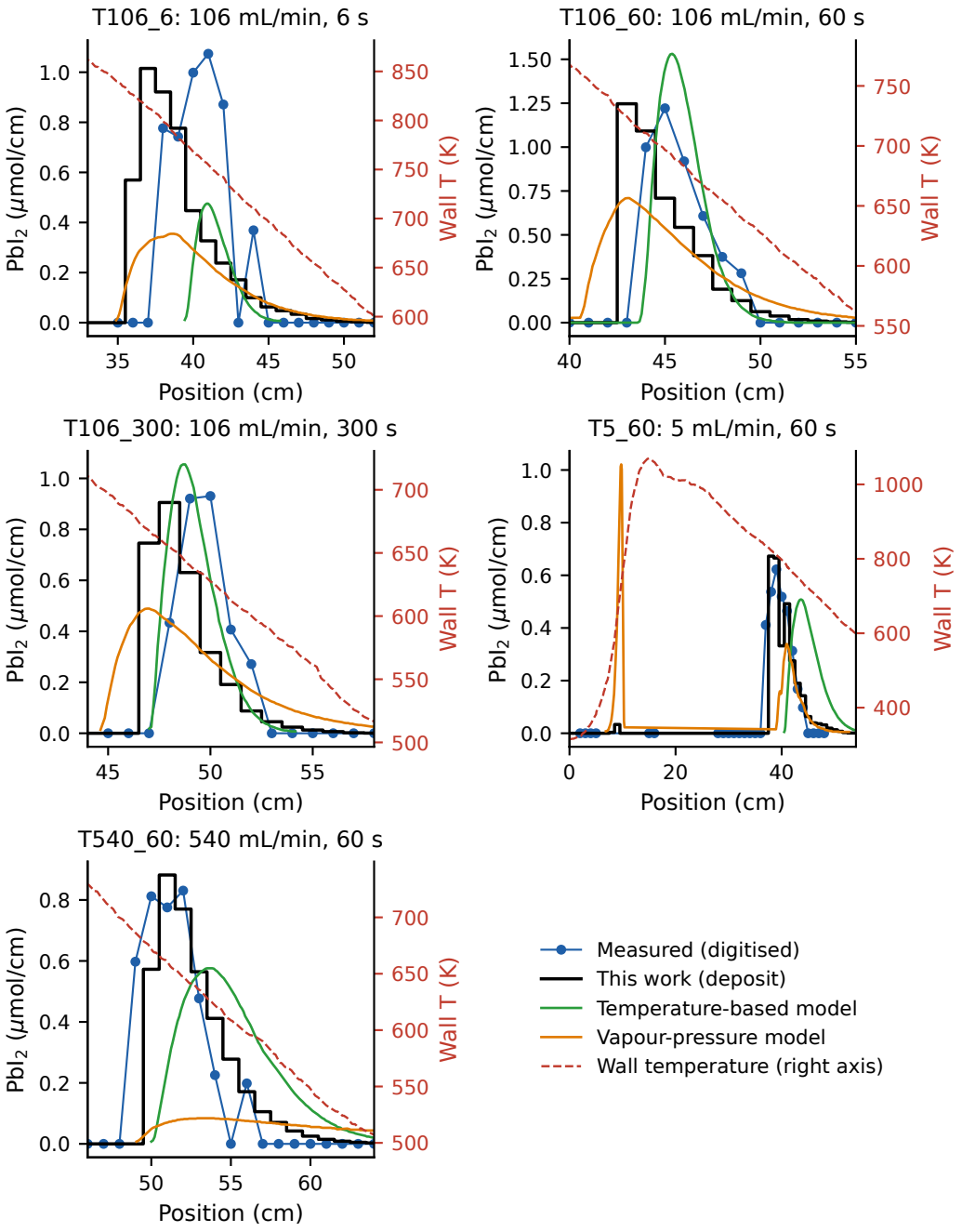


FIG. 5. Deposited PbI_2 for the five experiments of Lobresco et al.[6]: measured profiles, their temperature-based and vapor-pressure-based model results (all digitized from their Figs. 13–14 and adapted under CC BY 4.0), and the deposit of this work at the end of each experiment (1 cm bins, 22,896 cells). Dashed red lines: digitized wall temperature used as input (right axes).

B. Silica temperature-coordinate inconsistency

In the silica experiment the published peak labels do not match the published temperature curve. The scan places the PbI_2 and BiI_3 maxima at 51 and 59 cm, labelled 306±24 and 127±39 °C. At those positions the digitized temperature curve gives approximately 369 and

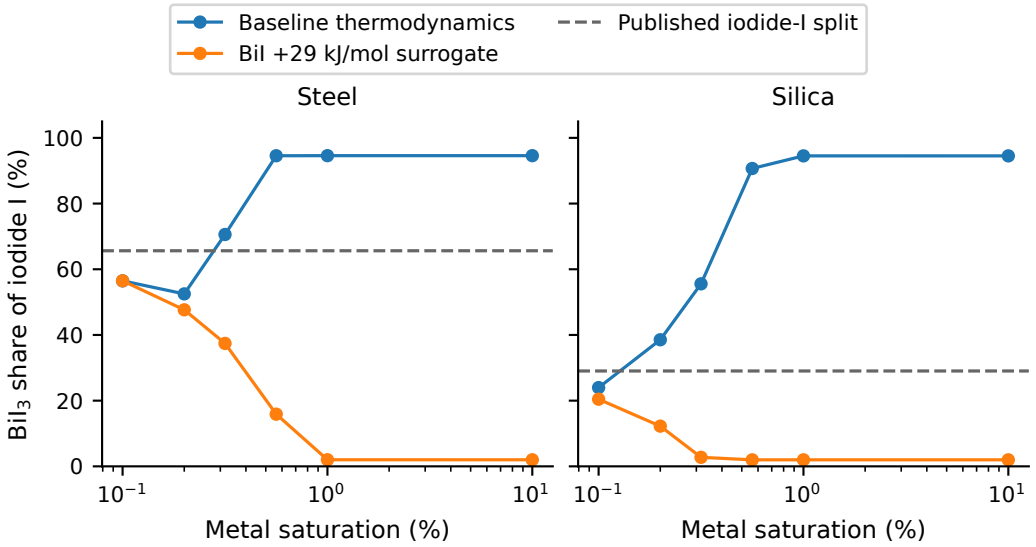


FIG. 6. BiI_3 share of deposited iodide iodine (eq. (20)) against source loading for the baseline and the +29 kJ/mol BiI thermodynamics, for steel and silica LBE. Symbols denote completed Q2 cases; segments only connect those points. Dashed grey lines show the published BiI_3 share after excluding KI[1]. The +29 kJ/mol branch is a surrogate perturbation, not a second qualified thermodynamic dataset.

235 °C, 63 and 108 K above the labels; on the same curve, 306 and 127 °C occur at 55.096 and 63.653 cm. The discrepancy exceeds the nominal raster temperature resolution, so the published inputs contain a temperature-coordinate inconsistency; they do not say which measurement, annotation or registration is in error.

Two diagnostic mappings shift only the silica temperature curve upstream, by 4.096 or 4.653 cm, so that one of these curve crossings aligns with the corresponding labelled position: $T_\delta(x) = T_0(x + \delta)$ for $\delta > 0$, with $\Delta x_T = -\delta$. The iodine scan stays fixed. The shifts are derived from the published curve and labels, not optimised against the CFD profiles.

Table VI and fig. 7 show that the shifts move the Q1 maxima from 56/63 cm to 51/59 or 51/58 cm and raise the normalized overlap from 6.13% to 77.80% or 73.26%. The low-loading Q2 variants place the maxima at 51/59 or 51/58 cm, with overlaps of 77.88% and 75.54%. On the 18,432-cell upwind mesh the order of the two shifts was the opposite (75.29% and 81.00% for Q1; 79.16% and 83.33% for Q2), so the overlap does not rank the two hypotheses robustly. The two shifts also give different peak temperatures, and no single mapping reconciles both labels. The shifted low-loading Q2 cases keep a BiI_3 share near 23.97%, about five percentage points below the published iodide-only split, so correct peak positions do not ensure the correct partition.

These results demonstrate the inconsistency and show that a plausible coordinate change has a larger effect on the computed agreement than the tested source-position or time-step changes. They do not resolve the inconsistency: both shifts use labels from the same experiment, neither is confirmed, and the improved agreement is conditional on an unconfirmed hypothesis. Resolving it requires the original thermocouple profile and its coordinate registration.

C. Transport, kinetics, flow reference and flow stop

Table VII reports, for each tested category, the range of iodine profile changes E_{L1} (eq. (16)) from the designated baseline. These are responses to finite hypothetical changes, not statistical confidence limits. The source-loading row uses its 1% reference case. The

TABLE VI. Silica temperature-coordinate hypotheses: computed peak positions and agreement with the fixed experimental scan for the original and two shifted temperature curves. Shifts and peak positions are in cm; raw L1 (L_{raw}) is in percentage points and overlap in percent. Q2 uses baseline thermodynamics and 0.1% metal saturation.

Source	Δx_T	$x_{\mathrm{PbI_2}}$	$x_{\mathrm{BiI_3}}$	Raw L1	Overlap
Q1	0.00	56	63	183.84	6.13
Q1	-4.10	51	59	40.17	77.80
Q1	-4.65	51	58	51.44	73.26
Q2	0.00	55	63	183.84	6.13
Q2	-4.10	51	59	37.40	77.88
Q2	-4.65	51	58	47.21	75.54

TABLE VII. Range of the baseline-relative iodine profile change E_{L1} (percent) within each completed sensitivity category. The flow-stop row compares 80 s inventories with their 60 s restart state; the other rows compare 60 s inventories. The silica temperature-coordinate row uses the production cases (41,472 cells, limited-linear advection); all other rows are upwind follow-up cases on 18,432 cells compared with their own upwind baselines. Case counts and variations are listed in Table S1 of the supplementary material.

Assumption	Minimum	Maximum
Gas diffusion	0.15	14.20
Iodide accommodation	0.25	4.21
Source position	0.08	0.19
Q2 metal loading	0.00	112.14
Flow reference	5.09	10.93
Silica temperature coordinates	187.10	193.96
Flow-stop relaxation	1.10	4.63

temperature-coordinate row measures changes from the original silica input and can approach 200% when the profile supports move apart.

Diffusion alternatives change profiles by up to 14.20%. A 273.15 K flow reference raises the inferred helium mass flow by 9.15% relative to 298.15 K and changes profiles by up to 10.93%. Iodide accommodation changes give up to 4.21%, whereas moving the prescribed source region by ±1.5 cm gives at most 0.19%, a statement about the empty-tube source model that cannot rule out an effect of a real boat on the flow. Fig. S1 of the supplementary material shows the maxima. They are differences between upwind cases on the 18,432-cell mesh, which itself differs from the production scheme by 12–14% for the two cases of the scheme study (section V); how much of that error cancels in the differences has not been quantified, so effects of comparable size are not ranked precisely. The tube-scale Damköhler numbers of section IV B and the small accommodation response, at most 0.41% for $\alpha = 0.1$ and 4.21% for $\alpha = 0.01$, against 3.83–14.20% for halving or doubling D, are consistent with predominantly diffusion-controlled exchange in the tested model. This interpretation combines a tube-scale estimate with sensitivities from the upwind campaign; it does not establish a resolved transition between kinetic and diffusion control. The wall temperature also changes the equilibrium pressure, so its effect cannot be reduced to a change in transport resistance alone.

The four fixed-temperature flow-stop cases change the deposited iodine profiles by 1.10–4.63% over 20 s, with centroid shifts of at most 0.0422 cm. Residual gas can therefore alter wall inventories after release stops, but because these runs neither cool the column nor flush it, they do not predict the end-of-experiment redistribution.

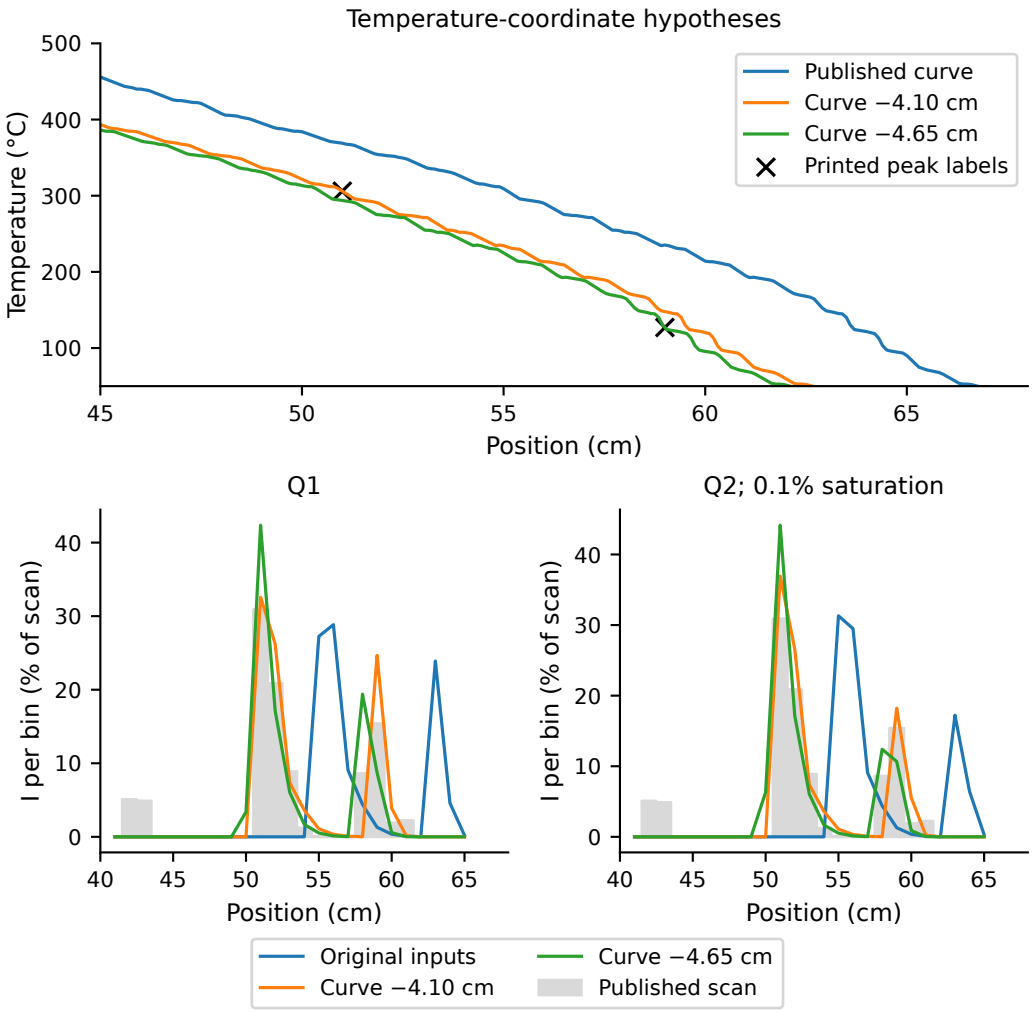


FIG. 7. Input-derived silica mappings and their consequences. Top: the original reconstructed temperature curve, the two shifts and the printed peak labels. Bottom: common-scan-normalized iodine profiles for Q1 and low-loading Q2. The improved agreement is conditional on an unconfirmed coordinate hypothesis. Published data are adapted from Liu et al.[1].

VIII. DISCUSSION: WHAT THE EVIDENCE SUPPORTS

A. Conservation, resolution and experimental agreement

The results separate four requirements that a deposition calculation must satisfy. Element balance tests the consistency of the discrete transport–equilibrium–wall coupling; the Graetz–Robin test verifies the wall discretization against a reference solution; resolution studies test how much the computed observables depend on numerical choices; and experimental comparison tests the combined consequences of the numerical model, its physical assumptions and its input data. Passing the first two does not establish the others.

Here the raw material residues are at round-off and the half-cell wall law converges at second order, yet local profile shapes remain resolution-sensitive while integrals are stable, and nearly equal bins can shift a reported peak temperature by tens of kelvin. Profile differences, centroids and deposited amounts should therefore accompany peak comparisons. The physical-input responses are finite responses over selected cases; they cannot be added

in quadrature or read as a universal parameter ranking, and several are comparable in magnitude to the unresolved spatial differences.

B. What the experimental comparisons can identify

Q1 uses deposit-derived compound masses for LBE, so agreement in the iodide split partly reflects information supplied to the calculation. Q2 lets the split emerge but combines a deposit-derived iodine loading with an unmeasured metal saturation; the computed split responds strongly to both the BiI thermodynamic input and the source loading, so one observed split constrains several coupled assumptions rather than identifying one of them. The silica scan and its printed temperatures are mutually inconsistent, and the two diagnostic shifts expose rather than resolve this. The untuned Lobresco comparison gives onset temperatures closer to the measurements than both published models, but with different carrier, vapor-pressure and source assumptions, a rule-dependent T5_60 value and profiles displaced upstream; it is encouraging for the transport and wall law, not a demonstration of them. The practical contribution is therefore a framework for testing explicit input hypotheses while retaining conservation and exposing profile discrepancies. Fitting an interfacial coefficient to a single peak under these conditions could compensate for an error elsewhere in the model.

C. Limitations

The limitations are collected here and referred to elsewhere.

- *Physics not represented:* a frozen, trace-loaded carrier with the wall temperature imposed over each cross-section, so no buoyancy-driven secondary flow (possibly significant locally in a horizontal column; orientation unknown), no conjugate heat transfer and no non-dilute effects near a source; no aerosols, nucleation, Soret transport, KI or water/oxygen chemistry, substrate reactions or boat obstruction; instantaneous ideal-gas equilibrium and an irreversible BiI wall channel in place of unmeasured kinetics.
- *Inputs:* digitized rather than tabulated wall temperatures; an unconfirmed coordinate origin and the silica inconsistency; assumed controller reference conditions and column pressure; deposit-conditioned source masses and an unmeasured metal saturation; BiI and constructed BiI_3(l) thermodynamic records that need independent qualification.
- *Protocol:* 60 s source-driven runs at the three-hour mean release rate, flow-stop runs at fixed temperature, and no simulation of the cooling or withdrawal at the end of any experiment.
- *Numerics:* spatial qualification beyond one refined case, a consistent observed order for the iodine profiles, the follow-up sensitivities with the production scheme, a verification of the nonlinear HKS law, and the full-duration shifted-BiI corrector comparison are outstanding.

Each item addresses an ambiguity demonstrated by the completed cases. The most direct experimental constraints would be the original thermocouple and scan coordinates, controller reference conditions, an independently measured release history and the column orientation and pressure. None of the omitted effects can be inferred from the present sensitivity ranges.

IX. CONCLUSIONS

We have applied a conservative finite-volume treatment of nine Pb–Bi–I gases and four shared condensed wall inventories, with element-conserving gas equilibrium and reversible

face-flux wall exchange, to 15 baseline and 64 follow-up cases, 19 production re-runs and a scheme, mesh and iteration study. The columns are laminar helium flows (Re ≈ 0.7–3) in tubes with a steep axial wall-temperature gradient, where axial advection, radial diffusion and the interfacial exchange law together set the deposition profile. The worst raw element residue is 8.85×10^{-13} of supplied material in the follow-up cases and 1.41×10^{-12} in the production cases; conservation is a numerical property of the model, not evidence that all physical pathways are represented. On a Graetz–Robin problem the half-cell wall law converges at second order to the reference decay rate, whereas applying the kinetic law to the cell-center value is first order; this verifies the wall discretization, not the convergence of the iodine calculations. Large tube-scale wall Damköhler numbers and the stronger response to diffusivity than to accommodation in the upwind sensitivity campaign are consistent with predominantly diffusion-controlled exchange. A resolved regime transition and the sensitivity ranking on the production scheme remain to be established.

Integral results are much less sensitive to resolution than local profiles. Halving the time step changes iodine profiles by at most 0.323%, and eight instead of three species correctors by at most 0.021%. The spatial error is larger and is dominated by the numerical diffusion of first-order upwind advection, which differs from the limited-linear scheme by 7.7–20.8% at equal mesh in the most sensitive case, whereas two bounded schemes differ by at most 2.8%. With limited-linear advection that case changes by 1.98% between 41,472 and 93,312 cells, but the apparent orders are inconsistent, so we quote this change rather than a formal grid convergence index. Closely competing 1 cm bins can shift a reported peak temperature by tens of kelvin, so peaks are reported with their smoothed maximum and centroid temperature as well.

The computed iodide partition depends strongly on thermodynamic input. In the refined steel case, a +29 kJ/mol shift of the BiI gas Gibbs energy changes the BiI_3 iodine share from 94.58% to 2.01%, and a comparable separation persists at every tested resolution. This shows how much the prediction depends on the BiI data; it is neither an uncertainty interval nor a criterion for choosing a dataset. For silica, temperature-coordinate shifts derived from the published curve and labels reduce 4–5 cm position discrepancies to 0–1 cm. This demonstrates an inconsistency in the published coordinates without resolving it; the overlap ranking of the two shifts even reverses between advection schemes.

On five single-species PbI_2 experiments of Lobresco et al., the untuned model gives deposition-onset temperatures with an RMS deviation of 18 K, against 35 and 35 K for the two models of that study. The result is promising but conditional: it depends on applying the onset rule to the main deposit (from the inlet, the RMS would be 135 K), the near-source mole fraction of 0.33–0.36 violates the dilute frozen-carrier assumption, and the release history is unconstrained below 6 s. Because carrier, vapor-pressure and source assumptions also differ from the published models, the improvement cannot yet be attributed to the wall treatment, and the computed peaks lie 1–4 cm upstream with summed profile differences of 0.37–0.89 of the sample amount.

The evidence supports a conservative method with a verified wall discretization and an explicit assessment of uncertain inputs, not independent predictive validation. The requirements for the latter are listed in section VIII C.

SUPPLEMENTARY MATERIAL

See the supplementary material for the case inventory, the detailed material-ledger and solver diagnostics, the scalar mesh checks of the upwind campaign, the comparison-bin-width analysis, the property and thermodynamic records, and the complete Graetz–Robin verification table.

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AUTHOR DECLARATIONS

Conflict of Interest

[AUTHOR TO CONFIRM: The authors have no conflicts to disclose.]

Author Contributions

Jan Kren: [AUTHOR TO CONFIRM: CRediT roles]. **Bojan Ničeno**: [AUTHOR TO CONFIRM: CRediT roles].

DATA AVAILABILITY

The solver, the benchmark preparation scripts and the data that support the findings of this study are openly available in the LESTO repository at `https://github.com/JanKren/LESTO`, [AUTHOR TO CONFIRM: archived release or DOI, and that the branch holding these files is public]. These files comprise the completed baseline, follow-up, production, scheme and mesh, and iteration-check metrics, the Graetz–Robin verification records, saved axial profiles, generated tables, figure-generation code and SHA-256 provenance records. The experimental curves were digitized from Liu et al.[1]; they reconstruct published figures, are not author-supplied measurements, and are adapted under the source paper's CC BY 4.0 license. The measured profiles, wall temperatures and model curves of Lobresco et al.[6] were digitized in the same way and are adapted under that paper's CC BY 4.0 license; the case set-up and final profiles of our comparison runs are included. Some thermodynamic coefficient files are external inputs that are not redistributed; their local identifiers and the provenance of the derived tables are recorded with the benchmark. Reproducing the CFD calculations from the coefficients requires access to those inputs, whereas regenerating the figures from the saved results does not.

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